

$$1) (a+ib)^2 = 3+4i$$

$$a^2 - 2abi - b^2 = 3+4i$$

$$\begin{cases} b = \frac{2}{a} \\ a^2 - \left(\frac{2}{a}\right)^2 = 3+4i \end{cases}$$

$$\begin{cases} b = \frac{2}{a} \\ (a^2)^2 - 3a^2 - 4 = 0 \end{cases}$$

$$(a^2-4)(a^2+1) = 0$$

$$a^2 - 4 = 0$$

$$a^2 = 4$$

$$a = \pm 2$$

$$a = 2 \Rightarrow b = 1$$

$$a = -2 \Rightarrow b = -1$$

Rez.

$$z^2 = 3+4i = \sqrt{5}(\cos \theta + i \sin \theta)$$

$$\theta = \arctg \frac{4}{3}$$

$$z_k = \sqrt[2]{5} \left(\cos \frac{\theta + 2k\pi}{2} + i \sin \frac{\theta + 2k\pi}{2} \right)$$

$$k = 0, 1$$

$$i\pi + \varepsilon = s(d + \rho) \quad (1)$$

$$2) f: C \rightarrow \mathbb{C}$$

$$i\pi + \varepsilon = s d - i d s - s \rho$$

$$f(x + iy) = (x \cos y - y \sin y) + i(y \cos y + x \sin y)$$

$$N.V. \quad \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$$

(u, v) nu verifica (C.R.)

3)

$$f(z) = \frac{z}{z^3 - z^2 - z + 1}$$

$$, \text{res}(f, 1) = ?$$

$$f(z) = \frac{z}{(z^2 - 1)(z - 1)}$$

$$z^3 - z^2 - z + 1 = (z - 1)^2(z + 1)$$

$$(z - 1)^2(z + 1) = 0$$

$$s \pm = 0$$

$$\frac{z}{(z - 1)^2(z + 1)}$$

$$i + s = \frac{1}{2}$$

$$i + s = -\frac{1}{2}$$

1 = pol ordin 2

$$d = d \quad c = s = 0$$

$$d = d \quad c = s = 0$$

$$\text{res}(f, 1) = \frac{1}{1!} \cdot \frac{d}{dz} \left(\frac{z}{z + 1} \right)$$

$$z \quad z - 2$$

$$s \quad \text{res}$$

$$z \quad \frac{z + 1}{(z + 1)^2}$$

$$z \quad \frac{z}{(z + 1)^2}$$

$$z \quad \frac{1}{4}$$

$$\text{res}(f^2, 0)$$

$$\frac{1}{z} + a = f(z) \text{ then } \text{res}(f, 0) = \frac{1}{s} = (f) \rightarrow (2)$$

$$f^2 = \frac{1}{z^2} + 2a \frac{1}{z} + a^2 = f \left(\frac{1}{(1-s)(1+s)} \right) = \frac{1}{(1-s)s} = (f)f$$

$$[\text{res}(f, 0)]^2 = 1$$

$$\text{res}(f^2, 0) = 2a. \text{ Alegem } a = \frac{1}{2}$$

$$f(z) = \frac{1}{z} \cdot \sum_{n=0}^{\infty} \left(\frac{s}{1+s} \right)^n \cdot \frac{1}{s} = \frac{1}{s(1-s)} = \frac{1}{s} \cdot \frac{1}{1-s}$$

$$f(z) = \frac{1}{z}$$

$$1 > |s|$$

$$5) f(-z) = \sum_{n=-\infty}^{\infty} a_n (-z)^n$$

$$\sum_{n=-\infty}^{\infty} a_n z^n = \sum_{n=-\infty}^{\infty} a_n (-z)^n$$

$$\sum_{n=-\infty}^{\infty} -a_n z^n = \sum_{n=-\infty}^{\infty} (-1)^n a_n z^n$$

$$\left. \begin{aligned} f \text{ ind } n \in \mathbb{Z} \cdot a_n \cdot i \pi s &= sb \frac{(s)^n}{(s)^n} \\ -a_n z (-1)^n a_n, \forall n \in \mathbb{Z} \end{aligned} \right\} = \frac{1+s}{1+s+s} \quad (3)$$

$$a_n [(-1) - (-1)^n] = 0, \forall n \in \mathbb{Z} \left(\begin{aligned} \text{deg. } n &= \text{tmi mib} \\ \text{deg. } n &= \text{tmi mib} \end{aligned} \right)$$

$$= \begin{cases} 0 & \text{dacă } n \text{ impar} \\ -2 & \text{dacă } n \text{ par} \end{cases}$$

$$a_m = 0 \quad \forall m \text{ par.}$$

$$(0, f)_{\text{ord}}$$

$$2) \quad f(z) = \frac{1}{z^4 z^2} = (0, f) \rightarrow \text{poli } (z) f = 0 + \frac{1}{z}$$

$$f(z) = \frac{1}{z^2(z^2+1)} = \frac{1}{z^2(z+i)(z-i)} + \frac{1}{z^2} = \frac{1}{z^2} + \frac{1}{5} \frac{1}{z+i} + \frac{1}{5} \frac{1}{z-i}$$

$$z=0 \text{ poli ord } 2$$

$$1 = [(0, f)_{\text{ord}}]$$

$$z \pm i \text{ poli ord } 1$$

$$= \frac{1}{z^2} \cdot \frac{1}{1-(-z)^2} = \frac{1}{z^2} \cdot \sum_{n=0}^{\infty} (-z^2)^n = \frac{1}{z^2} \cdot (1 - z^2 + z^4 - z^6 + \dots)$$

$\uparrow$
 $|z|^2 < 1$

$$= \frac{1}{z^2} - 1 + z^2 - z^4 + z^6 - \dots$$

$$\sum_{n=0}^{\infty} (-1)^n \cdot z^{2n-2}$$

$$\textcircled{3} \quad \int_{|z|=2} \frac{2z+1}{z^2+z+1} dz = \int \frac{f'(z)}{f(z)} dz = 2\pi i (\text{nr. zero lui } f)$$

$$\dim \text{int } C = \text{nr. poli} \quad \mathbb{C} \cong \mathbb{R}^2$$

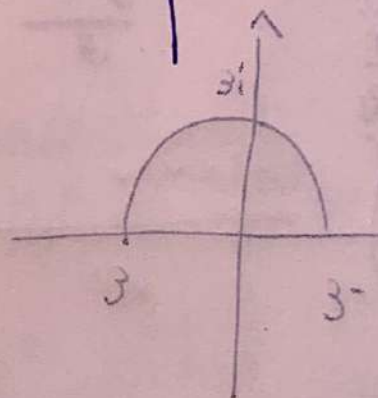
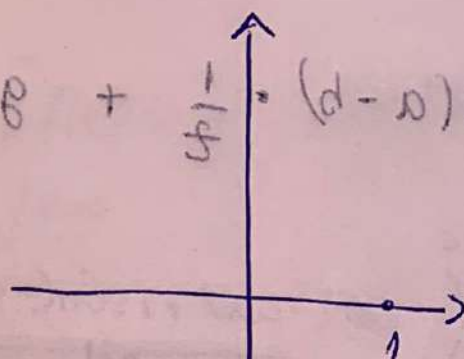
$$f(z) = 0 \Leftrightarrow z^2 - 1 = 0 \quad z_1 = 1 \quad z_2 = -1$$

$$z = \frac{-1 + i\sqrt{3}}{2}$$

$$| -\frac{1 - i\sqrt{3}}{2} - 1 | = | -\frac{3}{2} - i\frac{\sqrt{3}}{2} |$$

$$= \sqrt{\frac{9}{4} + \frac{3}{4}} = \frac{3}{2} \in \text{int } C$$

$$= \sqrt{3} < 2$$



$$f: \mathbb{C} \setminus \{0\} \rightarrow \mathbb{C}$$

$$f(z) = \frac{e^{iaz} + e^{ibz}}{z^2}$$

$$e^{iaz} = \sum_{n=0}^{\infty} \frac{(iaz)^n}{n!}$$

$$f(z) = \frac{1}{z^2} \left[\sum_{n=0}^{\infty} \frac{(iaz)^n}{n!} + \sum_{n=0}^{\infty} \frac{(ibz)^n}{n!} \right]$$

$$= \frac{1}{z^2} \left(1 + iaz - \frac{a^2 z^2}{2!} - \frac{i a^3 z^3}{3!} + \dots + 1 + ibz - \frac{b^2 z^2}{2!} - \frac{i b^3 z^3}{3!} + \dots \right)$$

$$\frac{1}{z^2} = i(a-b) \cdot \frac{1}{z} + \underbrace{\left(-\frac{a^2}{2!} + \frac{b^2}{2!} \right) + \left(-\frac{ia^3}{3!} + \frac{ib^3}{3!} \right) z}_{g(z)}$$

$$\frac{\partial \sqrt{1+i\sqrt{3}}}{\partial z}$$

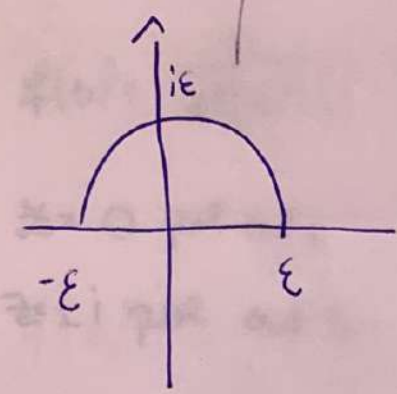
$$z \cdot i(a-b) \cdot \frac{1}{z} + g(z)$$

$$\left| \frac{\partial \sqrt{1+i\sqrt{3}}}{\partial z} \right| = \left| \frac{1}{2} \cdot \frac{1}{\sqrt{1+i\sqrt{3}}} \cdot i\sqrt{3} \right|$$

$$\lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\frac{\epsilon}{2} + \frac{\epsilon}{2} \right) = 1$$

$$s > \sqrt{3}$$

b)



$$f: C \setminus \{0\} \rightarrow \mathbb{C}$$

$$\lim_{\epsilon \rightarrow 0} \int_{\gamma_\epsilon} \frac{i(a-b)}{z} dz + \lim_{\epsilon \rightarrow 0} \int_{\gamma_\epsilon} g(z) dz = f(s)$$

$$\int_{\gamma_\epsilon} \frac{i(a-b)}{z} dz = i^2 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (a-b) dt = - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (a-b) dt$$

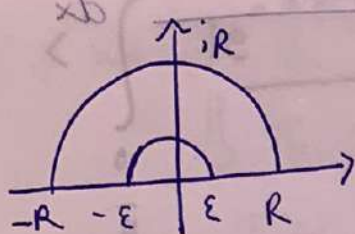
$$(a-b)\pi = \pi(a-b) = \frac{\pi}{s} = f(s)$$

$$\left| \int_{\gamma_\epsilon} g(z) dz \right| \leq \left(\max_{z \in \gamma_\epsilon} |g(z)| \right) \cdot \text{length}(\gamma_\epsilon)$$

$$\lim_{\epsilon \rightarrow 0} \left(\max_{z \in \gamma_\epsilon} |g(z)| \right) \cdot \text{length}(\gamma_\epsilon) = 0$$

$$f(2) = 0$$

$$c) \int_0^{\infty} \frac{\cos(ax) - \cos(bx)}{x^2} dx$$



Th. Cauchy $\int_{\delta_{\epsilon,R}} f(z) dz = 0$

$$\int_{\delta_{\epsilon,R}} f(z) dz = \int_{-R}^{-\epsilon} \frac{e^{iax} - e^{ibx}}{x^2} dx + \int_{\epsilon}^R \frac{e^{iax} - e^{ibx}}{x^2} dx + \int_{\epsilon}^R \frac{e^{ia(Re^{it})} - e^{ib(Re^{it})}}{i(Re^{it})^2} dt$$

$$+ \int_{-R}^{-\epsilon} \frac{e^{ia(Re^{it})} - e^{ib(Re^{it})}}{i(Re^{it})^2} dt$$

$$= \int_{-R}^{-\epsilon} \frac{\cos(ax) - \cos(bx) - i\sin(ax) + i\sin(bx)}{x^2} dx + \int_{\epsilon}^R \frac{\cos(ax) - \cos(bx) - i\sin(ax) + i\sin(bx)}{x^2} dx$$

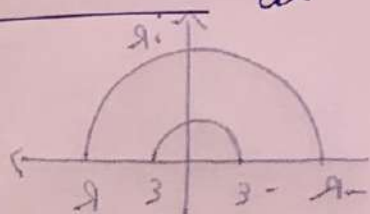
$$+ \int_{-R}^{-\epsilon} \frac{\cos(ax) + i\sin(ax) - \cos(bx) - i\sin(bx)}{x^2} dx + \int_{\epsilon}^R \frac{\cos(ax) + i\sin(ax) - \cos(bx) - i\sin(bx)}{x^2} dx$$

$$+ \int_{\epsilon}^R \frac{\cos(ax) + i\sin(ax) - \cos(bx) - i\sin(bx)}{x^2} dx$$

$$+ \int_{\delta\epsilon}^{\epsilon} f(z) dz$$

$$\left. \frac{(x_0) \cos(-bx) - (x_0) \cos(-ax)}{s x} \right\}$$

$$z = \int_{\delta\epsilon}^{\epsilon} \frac{\cos(-ax) + i \sin(-ax) - \cos(-bx) - i \sin(-bx)}{s(-x)^2} dx$$



$$+ \int_{\delta}^R \frac{\cos(ax) + \dots}{x^2} dx + \int_{\delta\epsilon}^{\epsilon} f(z) dz$$

$$+ \int_0^{\pi} \left[\frac{e^{iaR \cos t - aR \sin t} - e^{ibR \cos t - bR \sin t}}{R e^{it} x} \right] dt$$

$$z = \int_{\delta}^R \left(\frac{\cos(ax) - \cos(bx)}{x^2} + i \frac{\sin(ax) + \sin(bx)}{x^2} \right) dx$$

$$+ \int_{\delta}^R \left(\frac{\cos(ax) - \cos(bx)}{x^2} + i \frac{\sin(ax) - \sin(bx)}{x^2} \right) dx$$

$$+ \int_{\delta}^R f(z) dz + \int_0^{\pi} \left[\frac{e^{iaR \cos t - aR \sin t} - e^{ibR \cos t - bR \sin t}}{R e^{it} x} \right] dt$$

$$\left. \frac{(x_0) \cos(-bx) - (x_0) \cos(-ax)}{s x} \right\}$$

$$= 2 \int_{\epsilon}^R \frac{\cos(ax) - \cos(bx)}{x^2} dx + \int_{\epsilon \rightarrow 0} f(z) dz + \int_0^{\bar{u}} \frac{1}{Re^a} dt$$

$$< \int_0^{\bar{u}} \frac{|e^{-R \sin t + i a R \cos t} - e^{-b R \sin t + i b R \cos t}|}{R} dt \leq$$

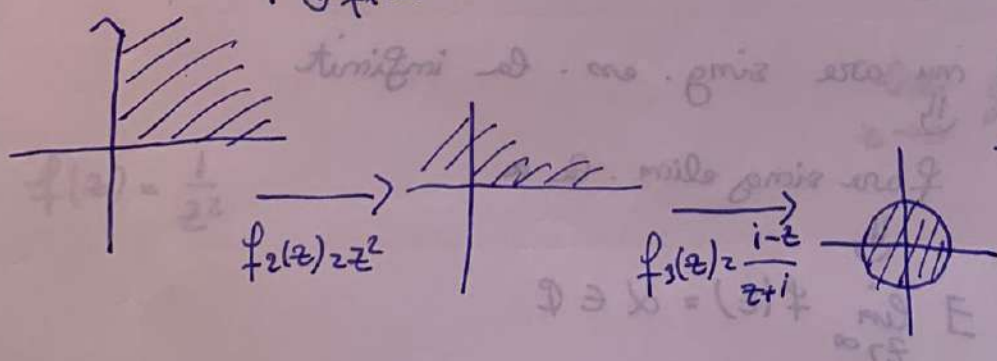
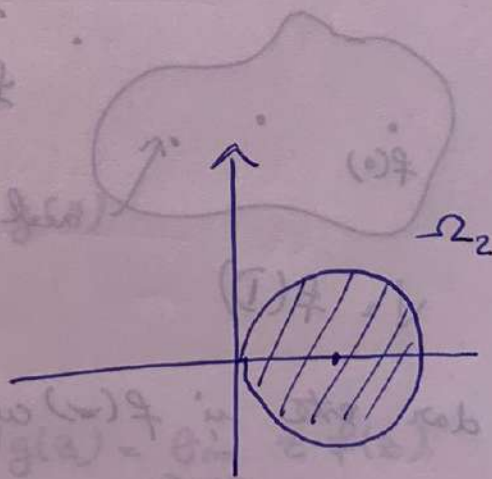
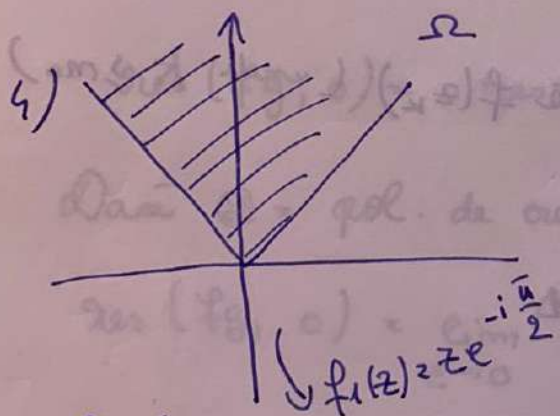
$$\int_0^{\bar{u}} \frac{|e^{-a R \sin t + i a R \cos t} - e^{-b R \sin t + i b R \cos t}|}{R} dt$$

$$\leq \int_0^{\bar{u}} \frac{2}{R} dt = \frac{2\bar{u}}{R} \xrightarrow{R \rightarrow \infty} 0$$

$$2\bar{u} + \bar{u}(a-b) + 0 = 0$$

$$\bar{u} = \frac{\pi}{2}(b-a)$$

e^z	$\frac{az+b}{az+a}$	rotati
$\log z$	$\frac{i-z}{i+z}$	translati
z^n	$i \cdot \frac{1-z}{1+z}$	$\frac{1}{2}(z + \frac{1}{z})$
$z^{\frac{1}{n}}$		



5) $f: \mathbb{C} \rightarrow \mathbb{C}$ oморfă + injectivă, atunci f

Nu poate avea sing. esențială la infinit

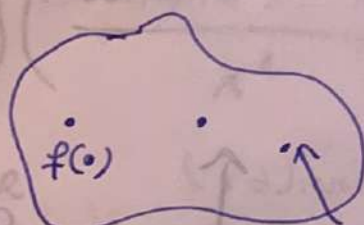
Rez: f oмор + inj $\Rightarrow f$ neconst
 $\Downarrow$ Th. apl. deschisă

deci $f(D) = V = \text{vec. deschisă a lui } f(0)$

PP. prim real. abs. că $\infty = \text{sing. deschisă}$

Th. C.W $\Rightarrow \exists (z_m)_{m \geq 1}, z_m \rightarrow \infty$ a.i. $f(z_m) \rightarrow f(0)$

$\Rightarrow f(z_m) \in V, \forall m \geq m_0$



$f(z_m)$

acest pct. este $f(z_k)$ (unde $k \geq m_0$)

$V = f(D)$

dar este și $f(w)$ cu $w \in D$

(b) (a) $\Rightarrow f$ nu are sing. es. la infinit
 $\Downarrow$

f are sing. elim. la ∞

$\Downarrow$
 $\exists \lim_{z \rightarrow \infty} f(z) = \alpha \in \mathbb{C}$

2) f mărginită pe $\mathbb{C}$

$\Downarrow$
 f const $\Rightarrow f$ nu e inj. $\mathcal{R}$

$0 = (0, f) \text{ cor}$

$1 = (0, f) \text{ cor}$

$\frac{1}{5} = (5, f)$

$5 = (5, f)$

impl

f are pol la ∞

$\Downarrow$ Th. 3.1. 15

$$f(z) = \frac{P(z)}{Q(z)}$$

P, Q - polinoame

$$(P, Q) = 1$$

f este pe $\mathbb{C} \Rightarrow Q$ nu se anulează

Q constantă

$$f(z) = \text{polinom} = R(z)$$

f inj

$\} \Rightarrow \cancel{f(z)} R$ de grad 1.